

# DR. AHMAD NAJMUDDIN IBRAHIM

Senior Lecturer (Pensyarah Universiti, DS13-A) & Robotics Researcher

Faculty of Manufacturing & Mechatronic Engineering Technology  
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## PROFESSIONAL PROFILE

Dr. Ahmad Najmuddin bin Ibrahim is a Senior Lecturer (Pensyarah Universiti, Grade DS13-A) at the Faculty of Manufacturing & Mechatronic Engineering Technology, Universiti Malaysia Pahang Al-Sultan Abdullah. His research focuses on robotics engineering, mobile robotics, and mechatronic systems, with particular expertise in autonomous wheeled robots for agricultural applications, rover locomotion in challenging terrains, and assistive grouser mechanisms for soft sand surfaces. He has published extensively in international journals and conferences and actively collaborates with Ibaraki University, Japan, while seeking partnerships with universities and industries worldwide.

## ACADEMIC QUALIFICATIONS

**Doctor of Engineering (D.Eng.) — Industrial Science (Robotics Engineering)** 2013 – 2017

*Ibaraki University, Hitachi, Japan*

- Research on assistive grouser mechanisms for wheeled rover locomotion on soft sand inclines.
- Developed prototype rovers tested on unconsolidated sandy terrain.

**Master of Engineering (M.Eng.) — Intelligent Systems Engineering** 2011 – 2013

*Ibaraki University, Hitachi, Japan*

- Research on prototype adaptable feet sole mechanism using vacuum-jamming mechanism.

**Bachelor of Engineering (B.Eng.) — Intelligent Systems Engineering** 2008 – 2011

*Ibaraki University, Hitachi, Japan*

## AREAS OF EXPERTISE

**Robotic Mechanical System Design** (*High expertise, 5+ years*): Rover design, grouser mechanisms, soft-soil locomotion, prototype development

**Mechanical-Electronics Systems Integration** (*High expertise, 5+ years*): Mechatronics, sensor integration, embedded systems, actuator control

**Discrete Element Modelling (DEM)** (*Moderate expertise, 1+ year*): Soil-wheel interaction simulation, terramechanics, granular material modelling

**Mobile Robotics and Autonomous Navigation** (*Expert*): Autonomous navigation, wheeled robots, agricultural robots, inspection robots

**Rover Locomotion in Soft Terrains** (*Expert*): Terrain-adaptive traction, unconsolidated sand and soil mobility

## CURRENT POSITIONS & RESPONSIBILITIES

**Senior Lecturer (Pensyarah Universiti, DS13-A)** May 2017 – Present

*Faculty of Manufacturing & Mechatronic Engineering Technology, UMPSA*

**Previous Positions:**

- **Head of Programme — Bachelor of Mechatronic Engineering Technology (Robotics)** January 2021 – January 2026

## TEACHING EXPERIENCE

**Core Courses Taught (2017 – Present):**

- Microcontroller System (BFM3333)
- Robotic Prototype Design (BFM4533)
- Robotics System Modelling (BTX2643)
- Capstone Project (BTI3623, BTT3623)

- Control System Engineering (BHM4103, BFF4103)
- Engineering Dynamics (BTM1223)
- Computer Programming (BFF2003, BHM2003)
- Industrial Automation (BHM4704)

## RESEARCH SUPERVISION

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### Doctoral (Ph.D.) Students:

- Nor Hazwani Binti Mohd Yusof — Design and Performance Analysis of Prototype Modular Wheeled Unmanned Ground Vehicle
- Mochamad Edoward Ramadhan — Outdoor Robot Wheel Design for Effective Use in Agriculture

### Master's (M.Sc.) Students:

- Ibrahim Belal Mogbhawry Mohamed — Enhancing Laser Simulator for Path Planning Processing
- Harun Dzulquornain Bin Idris — Development and Performance Evaluation of UWB+IMU Navigation
- Intan Nur Aqiella Binti Che Aziz — Mathematical Analysis of Wheel-Soft Soil Interaction
- Siti Suhaila Binti Sabarudin — 2D Simulation of Assistive Grouser Mechanism Using DEM

## RESEARCH PROJECTS

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### Principal Investigator:

#### Automated Cleaning and Monitoring of HVAC Duct Using Mobile Robot

2024 – 2026

Development of a tethered mobile robot platform for automated HVAC duct inspection and cleaning, integrating visual and environmental sensors for condition assessment.

- Grant reference: RDU243812, UIC240849

#### Grouser Effect Contact Modelling in Cohesive Unstructured Terrains

2023 – 2025

Investigation of grouser-soil interaction mechanisms in cohesive unstructured terrains using experimental and simulation approaches.

- Grant reference: RDU230135

### Co-Investigator:

#### DREBAR Education Robotic Module for ADTEC Kemaman

2024 – 2025

- Grant reference: CDU240136

#### Development of Remotely Underwater Cleaning Robot

2023 – 2024

- Grant reference: SPU230102

### Completed Research Projects (as Principal Investigator):

#### Development of Autonomous Mobile Robot for Optimum HVAC Duct Cleaning

2021 – 2022

- Grant reference: RDU212402, UIC210810

#### Projek Khas Robotik

2022

- Grant reference: SPU220101

#### Mathematical Analysis of Wheel-Soft Soil Interaction for Lightweight Vehicle Wheel with Variable Grouser Angle Mechanism

2019 – 2021

- Grant reference: RDU190115

#### Simulation of Assistive Grouser Mechanism for Use on Lightweight Wheeled Robot for Traversing on Unconsolidated Soft Sand Inclines

2019 – 2021

- Grant reference: PGRS190373

#### Design Strategy of Stuck Recovery for Wheel Embedded Under the Surface of an Unconsolidated Soft Sand Incline

2018 – 2020

- Grant reference: RDU180384

#### Simulation of Assistive Grouser Mechanism for Use on Lightweight Wheeled Robot for Traversing on Unconsolidated Soft Sand Inclines

2017 – 2019

- Grant reference: RDU1703191

## SELECTED PUBLICATIONS

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- Ibrahim, A.N., Aoshima, S., Shiroma, N., Fukuoka, Y. "The effect of assistive anchor-like grousers on wheeled rover performance over unconsolidated sandy dune inclines." *Sensors* 16, no. 9 (2016): 1507.
- Fukuoka, Y., Komatsu, R., Machii, K., Yokota, M., Tobe, M., Ibrahim, A.N., et al. "Pace running of a quadruped robot driven by pneumatic muscle actuators: an experimental study." *Applied Sciences* 12, no. 9 (2022): 4146.
- Fukuoka, Y., Oshino, K., Ibrahim, A.N. "Negotiating uneven terrain by a simple teleoperated tracked vehicle with internally movable center of gravity." *Applied Sciences* 12, no. 1 (2022): 525.
- Najmuddin, A., Fukuoka, Y., Ochiai, S. "Experimental development of stiffness adjustable foot sole for use by bipedal robots walking on uneven terrain." 2012 IEEE/SICE International Symposium on System Integration.

*Complete publication record: 27 peer-reviewed publications (2012–2024), including journal articles, conference papers, and book chapters.*

## UNIVERSITY SERVICE

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**Head of Programme — Bachelor of Mechatronic Engineering Technology (Robotics)** *Jan 2021 – Present*

**Advisor / Escorting Officer for Student Competitions:**

- National Instruments Autonomous Robotics Competition 2019 — Universiti Sains Malaysia, Nov 12–14, 2019
- Pertandingan Robocon Malaysia 2019 — Kementerian Pengajian Tinggi, UNITEN, Apr 4–7, 2019
- Pertandingan University Robot Competition 2019 — Universiti Malaysia Perlis, Kampus Pauh Putra, Sep 21–22, 2019
- Pertandingan University Robot Competition 2018 — Universiti Malaysia Perlis, Kampus Pauh Putra, Oct 12–13, 2018

## AWARDS & RECOGNITION

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**GOLD Medal — ITEX 2024** *2024*

*International Invention, Innovation & Technology Exhibition (ITEX)*

Awarded for the project "Remotely Underwater Cleaning Robot".

**SILVER Medal — ITEX 2023** *2023*

*International Invention, Innovation & Technology Exhibition (ITEX)*

Awarded for the project "Agronetics IoT Based Smart Farming System".

## RESEARCH INTERESTS

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**Physical AI for Robotics** (*Focus Area*): learning-based control, AI-driven adaptation to real-world physical environments

**Terrain-Adaptive Robot Mobility** (*Focus Area*): Grouser mechanisms, soft-soil terramechanics, wheeled rover locomotion, traction optimisation

**Field & Inspection Robotics** (*Focus Area*): HVAC duct inspection, confined-space navigation, tethered robotic systems

**Agricultural Robotics** (*Focus Area*): Autonomous field robots, pest deterrence, unstructured terrain navigation

**Embedded Systems & Control** (*Focus Area*): Microcontroller systems, PID control, sensor fusion, real-time embedded programming

## PROFESSIONAL AFFILIATIONS & CERTIFICATES

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Institute of Electrical and Electronics Engineers (IEEE) — International Professional Member, 2018–present

Board of Engineers Malaysia (BEM) — Graduate Engineer, 2018–present

Malaysian Board of Technologists (MBOT) — Graduate Technologist, 2018

## LANGUAGES

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Malay — Native speaker | English — Fluent | Japanese — Conversational